

A COMPUTABLE DERANDOMIZATION OF A THIN-SET CONSTRUCTION

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STATUS AND SOURCE QUESTION

Status: substantial partial result (full effective derandomization; the “natural” closed-form question remains).

Neuwirth asks on page 3 of [1] whether some “natural” set of integers is a $\Lambda(p)$ set for every finite p and fails the Rosenthal property. The preceding sentence emphasizes that the random construction provides no explicit procedure for building such a set.

- E is equidistributed in Hermann Weyl’s sense: save for $t \equiv 0 \pmod{2\pi}$, the successive means of $\{e^{int}\}_{n \in E}$ tend to 0, which is the mean of e^{it} over $[0, 2\pi[$. This implies that E is not a Rosenthal set by [18, Lemma 4].

Our random construction gives no hint for explicit procedures to build such integer sets. The question whether some “natural” set of integers is $\Lambda(p)$ for all p and fails the Rosenthal property remains open. Let us describe the paper briefly. Section 3 introduces the inquired notions and gives a survey of former and new results. As the right framework for this study appears to consist in the sequences of polynomial growth, we give them a precise meaning in Section 4, and show that they are nicely distributed among the intervals of the partition of $\mathbb{Z}$ defined by $\{\pm 2^{k!}\}$. Section 5 establishes an optimal criterion for the generic subset of a set with polynomial growth to be $\Lambda(p)$ for all p . Section 6 comes back to Bourgain’s proof in [3, Prop. 8.2(i)]: we simplify and strengthen it in order to investigate the generic subset of an equidistributed set.

Notation $\mathbb{T} = \{t \in \mathbb{C} : |t| = 1\}$ is the unit circle endowed with its Haar measure dm and $\mathbb{Z}$ its dual group of integers: for each $n \in \mathbb{Z}$, let $e_n(t) = t^n$. The cardinal of $E = \{n_k\} \subseteq \mathbb{Z}$ is written $|E|$. We denote by $c_0(\mathbb{T})$ the space of functions on $\mathbb{T}$ which are arbitrarily small outside finite sets; such functions necessarily have countable support.

The theorem below gives a parameter-free deterministic algorithm for such a set. Its output is computable, but its exhaustive finite searches are enormous and it is not a closed formula. Since “natural” has no formal definition, the result is deliberately not presented as a complete answer to that adjective.

DEFINITIONS AND SOURCE INGREDIENTS

For $E \subseteq \mathbb{Z}$ and $p > 0$, recall that E is a $\Lambda(p)$ set if, for some $0 < r < p$, the L^p and L^r norms are equivalent on trigonometric polynomials with spectrum in E . If $E = \{n_j\}_{j \geq 1}$ is ordered increasingly, it is *equidistributed* if

$$\frac{1}{m} \sum_{j=1}^m z^{n_j} \longrightarrow 0 \quad (z \in \mathbb{T} \setminus \{1\}).$$

Neuwirth records that every weakly equidistributed set fails the Rosenthal property [1, p. 3].

We use two further ingredients proved in the source.

(N1) For an integer $s \geq 2$, call a finite set F *s-independent* if there is no relation

$$\sum_{j=1}^m a_j x_j = 0, \quad 3 \leq m \leq 2s, \quad x_j \in F \text{ distinct}, \quad a_j \in \mathbb{Z} \setminus \{0\},$$

with $\sum_j a_j = 0$ and $\sum_j |a_j| \leq 2s$. If I_k form a Littlewood–Paley partition and $F_k \subseteq I_k$ are *s-independent*, then $\bigcup_k F_k$ is a $\Lambda(2s)$ set [1, Proposition 5.4].

(N2) If each point of a finite set of size q is selected independently with probability ℓ/q , then

$$(1) \quad \mathbf{P}\{\text{the selected set is } s\text{-dependent}\} \leq D_s \frac{\ell^{2s}}{q},$$

where one may take, for example, $D_s = (2s)(4s+1)^{2s}$; see [1, Lemma 5.5] and its proof.

We also use the complex Bernstein concentration inequality in [1, Lemma 6.1]: if independent centered variables satisfy $|X_j| \leq 1$ and $\sum_j \mathbf{E}|X_j|^2 \leq v$, then

$$(2) \quad \mathbf{P}\left\{\left|\sum_j X_j\right| \geq a\right\} < 4 \exp\left(-\frac{a^2}{4(v+a)}\right).$$

THE RESULT

Theorem 1. *There is a parameter-free computable set $E \subseteq \mathbb{N}$ such that*

- (i) *E is a $\Lambda(p)$ set for every $p < \infty$;*
- (ii) *E is equidistributed;*
- (iii) *consequently, E is not a Rosenthal set.*

In particular, there is an explicit deterministic procedure producing a set with the two opposed properties in Neuwirth’s question.

IDEA OF THE PROOF

Use the factorial annuli with endpoints $p_k = 2^{k!}$. In block k we ask for about $(k+2)!$ points. Two finite requirements are imposed. First, the block must have no short additive relations, with the forbidden length tending to infinity with k . This makes the union $\Lambda(p)$ for every p by the source’s Littlewood–Paley criterion. Second, every prefix of the block must have small centered exponential sums on a sufficiently fine grid of roots of unity. Bernstein’s inequality shows that the second condition fails with very small probability; Neuwirth’s relation count does the same for the first. Hence at least one subset of every block passes both finite tests, and choosing the lexicographically first one is a deterministic, terminating algorithm.

The prefix condition is essential. Endpoint discrepancy alone would not control averages while the enumeration is partway through a large block. The adaptive threshold used below yields relative error $O(k^{-1/2})$ uniformly at every cutoff in block k .

CONSTRUCTION

For $k \geq 4$, set

$$p_k = 2^{k!}, \quad B_k = \{p_{k-1} + 1, \dots, p_k\}, \quad q_k = |B_k|, \quad \ell_k = (k+2)!, \quad \delta_k = \frac{\ell_k}{q_k}, \quad L_k = k!.$$

Let

$$h_k = \max\left(\{1\} \cup \left\{2 \leq s \leq k : D_s \frac{\ell_k^{2s}}{q_k} \leq \frac{1}{4}\right\}\right).$$

Because $\log q_k \asymp k!$ while $\log(D_s \ell_k^{2s}) = O_s(k \log k)$ for each fixed s , we have

$$(3) \quad h_k \longrightarrow \infty.$$

Let Ω_k be the set of all $(4p_k)$ th roots of unity. For $A \subseteq B_k$, $1 \leq r \leq q_k$, and $z \in \mathbb{T}$, define

$$Z_{A,k,r}(z) = \sum_{m=1}^r (\mathbf{1}_A(p_{k-1} + m) - \delta_k) z^{p_{k-1}+m}, \quad v_{k,r} = r\delta_k,$$

and put

$$T_{k,r} = 32 \left(\sqrt{(v_{k,r} + 1)L_k} + L_k \right).$$

Call A *admissible* if

- (A1) A is h_k -independent when $h_k \geq 2$;
- (A2) $|Z_{A,k,r}(z)| \leq T_{k,r}$ for every $1 \leq r \leq q_k$ and every $z \in \Omega_k$.

Order subsets of B_k lexicographically by their indicator words, let E_k be the first admissible subset, and define

$$E = \bigcup_{k \geq 4} E_k.$$

PROOF OF THEOREM 1

1. Every block search terminates. Select the elements of B_k independently with probability δ_k . For a fixed prefix r and fixed $z \in \Omega_k$, the summands defining $Z_{A,k,r}(z)$ are independent, centered, bounded by 1, and have total variance at most $v_{k,r}$. If $b = \sqrt{(v_{k,r} + 1)L_k} + L_k$ and $a = 32b$, then

$$\frac{a^2}{4(v_{k,r} + a)} \geq 4L_k.$$

Indeed, $b^2 \geq v_{k,r}L_k$ and $b \geq L_k$, so $64b^2 \geq L_kv_{k,r} + 32L_kb$. Applying (2) and taking a union bound over at most p_k prefixes and $4p_k$ grid points gives

$$(4) \quad \mathbf{P}\{(\text{A2}) \text{ fails}\} \leq 16p_k^2 e^{-4L_k} = 16 \exp(-(4 - 2 \log 2)L_k) < \frac{1}{4}.$$

For $h_k \geq 2$, (1) and the definition of h_k give

$$\mathbf{P}\{(\text{A1}) \text{ fails}\} \leq \frac{1}{4};$$

for $h_k = 1$ there is no condition. Thus an admissible subset exists for every $k \geq 4$. The lexicographic search is finite and terminates.

All its tests are decidable. Condition (A1) searches finitely many integer relations. In (A2), the numbers involved are rational combinations of roots of unity and square roots of rationals, so comparison can be carried out exactly in finite algebraic extensions of $\mathbb{Q}$. Given $n \in \mathbb{N}$, one locates its unique block and performs this finite search. Therefore E is a computable set.

2. The $\Lambda(p)$ properties. The symmetric factorial annuli

$$[-p_k, -p_{k-1}) \cup (p_{k-1}, p_k]$$

form the gross Littlewood–Paley partition used in [1, §5]. Fix $s \geq 2$. By (3), $h_k \geq s$ for all sufficiently large k , so E_k is s -independent in all sufficiently large blocks. Ingredient (N1) shows that the corresponding tail of E is a $\Lambda(2s)$ set. Adding the finitely many earlier points preserves that property. As s is arbitrary (and lower exponents follow from the usual interpolation/inclusion property), E is $\Lambda(p)$ for every finite p .

3. Uniform discrepancy through partial blocks. A trigonometric polynomial of degree at most p_k has, by Bernstein’s inequality, supremum at most five times its maximum on Ω_k : choose a nearest grid point, use $\|P'\|_\infty \leq p_k \|P\|_\infty$, and note that the angular mesh error is at most $\pi/(4p_k)$. Hence (A2) implies

$$(5) \quad \|Z_{E_k, k, r}\|_\infty \leq 160 \left(\sqrt{(v_{k, r} + 1)k!} + k! \right) \quad (1 \leq r \leq q_k).$$

Define a deterministic weight $\delta_n = \delta_k$ for $n \in B_k$ and zero before B_4 . If $N = p_{k-1} + r \in B_k$, put

$$\sigma_N = \sum_{n \leq N} \delta_n = A_k + v_{k, r}, \quad A_k = \sum_{i=4}^{k-1} \ell_i,$$

and

$$D_N(z) = \sum_{n \leq N} (\mathbf{1}_E(n) - \delta_n) z^n.$$

At the end of block i , (5) is $O(i!)$; consequently the sum of all completed-block errors before block k is $O(k!)$. Also $A_k \geq \ell_{k-1} = (k+1)!$. From (5), uniformly for $0 \leq v = v_{k, r} \leq \ell_k$,

$$(6) \quad \begin{aligned} \frac{\|D_N\|_\infty}{\sigma_N} &\leq \frac{O(k!)}{A_k + v} + 160 \frac{\sqrt{(v+1)k!} + k!}{A_k + v} \\ &= O(k^{-1/2}) \rightarrow 0. \end{aligned}$$

For the square-root term, use $v+1 \leq 2(A_k + v)$ and $A_k \geq (k+1)k!$.

Taking $z = 1$ in (6) gives

$$(7) \quad |E \cap [1, N]| = \sigma_N + D_N(1) \sim \sigma_N.$$

In particular, E is infinite.

4. Equidistribution and failure of Rosenthal. The block weights $\delta_k = \ell_k/q_k$ are eventually decreasing. Fix $z \in \mathbb{T} \setminus \{1\}$. The geometric partial sums $S_M(z) = \sum_{n \leq M} z^n$ are bounded. Abel summation, applied after the irrelevant finite initial segment, therefore shows that

$$(8) \quad W_N(z) := \sum_{n \leq N} \delta_n z^n = O_z(1).$$

Since $\sigma_N \rightarrow \infty$, equations (6), (7), and (8) give

$$\frac{1}{|E \cap [1, N]|} \sum_{n \in E \cap [1, N]} z^n = \frac{W_N(z) + D_N(z)}{\sigma_N + D_N(1)} \rightarrow 0.$$

This holds at every cutoff N , including cutoffs inside a block. Thus E is equidistributed. Neuwirth’s cited Lust–Piquard criterion now implies that $C_E(\mathbb{T})$ contains a copy of c_0 , and E is not a Rosenthal set. This proves the theorem. $\square$

VERIFICATION NOTES AND LIMITATIONS

- The simultaneous block requirements are compatible because their two failure probabilities sum to less than 1, not because either event is assumed independent of the other.
- The grid test includes $z = 1$, so the same discrepancy estimate controls the cardinality denominator. It also includes every prefix in a block; this prevents an endpoint-only gap.
- The construction is canonical once the displayed parameters and the lexicographic order are fixed. It is computable but not computationally feasible: block sizes are of order $2^{k!}$.
- This packet does not claim that a brute-force lexicographic object is a “natural” closed-form set. It fully answers the precise effective procedure version suggested by the sentence immediately preceding the source question.

NOVELTY CHECK AND HUMAN-REVIEW RECOMMENDATION

The bounded search on 27 August 2026 covered the four run indexes and web/ arXiv queries combining “explicit”, “computable”, “deterministic”, “equidistributed”, “ $\Lambda(p)$ for all p ”, and “Rosen-thal”. It found Neuwirth’s 1999 article (uploaded as arXiv:2603.28227) and the later random thin-set constructions [2, 3], but no matching computable derandomization. This was not an exhaustive citation-graph review; novelty confidence is therefore medium.

Human review should focus on (a) the exact algebraic decidability statement for the finite grid test, (b) the factorial estimate in (6), and (c) whether the community would regard effective derandomization as addressing enough of the word “natural” to merit a stronger status. The present recommendation is to retain **partial**, even though Theorem 1 itself is complete.

REFERENCES

- [1] S. Neuwirth, *Two random constructions inside lacunary sets*, Ann. Inst. Fourier (Grenoble) **49** (1999), no. 6, 1853–1867; arXiv:2603.28227, <https://arxiv.org/abs/2603.28227>.
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